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## **Rockhampton Sports Precinct Stage 1 – Rockhampton Netball Centre**

# **CIVIL AND FIELD OF PLAY TECHNICAL SPECIFICATION**

Revision: T1  
Date: 17/06/2026  
By: ICC  
Approved: JZV  
Reference: SE\_12306\_SPC001

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## Attachments

CMDG Construction Specifications (current versions)

# 1. Introduction

## 1.1 General

### Project particulars

- (a) This Specification has been developed specifically for the civil and field of play works on the following project / site:
- i. Project Name: Rockhampton Sports Precinct Stage 1 – Netball Centre
  - ii. Location: 700 Yaamba Road, Norman Gardens, QLD, 4701, Australia
  - iii. Principal: Rockhampton Regional Council
- (b) This document, the design drawings and all other referenced documentation describe the scope of works, minimum requirements, and assessment criteria.

### Contract documents

- (c) This Specification is to be read in conjunction with:
- i. General conditions of contract
  - ii. Special conditions of contract and supplementary special conditions of contract
  - iii. Contract preliminaries
  - iv. Contract drawings
  - v. Any other referenced documentation.

## 1.2 Scope of Works

### Quality management system

- (a) The Contractor shall implement a quality management system for all works under the Contract in accordance with *Section 3 Specification Preliminaries*.

### Works to be performed

- (b) Unless noted otherwise, provide all materials, labour, approvals, permits, verification, and testing as necessary to deliver the works in accordance with the drawings and specifications.
- (c) Works to be performed under this Specification include:
- i. Site clearing
  - ii. Bulk earthworks and in-situ stabilisation
  - iii. Construction of stormwater drainage
  - iv. Construction of concrete kerbs and pavements
  - v. Installation of fencing and gates
  - vi. Installation of pavement markings, signs, bollards, and the like
  - vii. Installation of specialist synthetic sports surface including line marking
  - viii. Supply and installation of sports equipment
  - ix. Supply of maintenance and operation manuals, as-built drawings
  - x. Defect rectification (during defined period)
  - xi. Any other civil work necessary to construct the works as documented.

## 2. Reference Documentation

### 2.1 General

#### Order of precedence

- (a) All works shall comply with the requirements of:
  - i. This Specification and Contract drawings
  - ii. Capricorn Municipal Development Guidelines
    - i. Design and Construction Specifications
    - ii. Standard Drawings
  - iii. Department of Transport and Main Roads Specifications and Standard Drawings
  - iv. Australian Standards (as applicable and/or referenced)
- (b) Applicable Australian Standards are referenced within the relevant work element sections.
- (c) Where requirements differ between referenced documents, the more onerous specification/ requirement applies unless otherwise directed by the Superintendent.

#### Contract drawings

- (d) Contract drawings (also called design drawings or the drawings) for this Specification are those shown in the Drawing List on Drawing Number SE\_12306\_F1000.

#### Status

- (e) The information contained in the drawings and specifications may also be subject to certification or approval by statutory and regulatory authorities.

### 2.2 Legislation and Standards

#### General

- (a) All works shall be carried out by suitably qualified tradespeople to a high standard of workmanship.
- (b) All works shall conform to this Specification and the relevant legislations and standards referenced or as otherwise applicable.
- (c) Any works carried out not confirming to these criteria shall be deemed as non-conforming and rectification works required to achieve this standard shall be the responsibility of the Contractor at no extra cost to the Principal.

## Legislation

- (d) The Contractor must carry out all the works in accordance with all relevant legislative provisions and standards including, but not limited to:
- i. Work Health and Safety Act 2011
  - ii. Environmental Protection Act 1994
  - iii. Local Government Act 2009
  - iv. Transport Infrastructure Act 1994
  - v. All relevant Australian Standards and Codes of Practice
  - vi. All other relevant State and Federal Acts and Regulations
  - vii. All Local Laws and relevant policies

## Standards

- (e) All works shall adhere to the relevant Australian Standards, referenced within each section of this Specification.
- (f) Additionally, all components of the Netball facilities shall adhere to the following requirements:
- i. International Netball Federation (INF) Facilities and Equipment Requirements
  - ii. AS/NZ 4586-2013 Slip Resistance Classifications of New Pedestrian Surface Materials
  - iii. AS/NZ 4633-2013 Slip Resistance Classification of Existing Pedestrian Surface Materials

## 2.3 Existing Site Information

### General

- (a) All existing site information is provided for information only, unless noted otherwise.
- (b) The Contractor must confirm the completeness and accuracy of all such information prior to commencing works.

### Discrepancies

#### HOLD POINT

- (c) Any discrepancies between the information provided in the specifications and drawings with that encountered on site must be reported to the Superintendent without delay.

HOLD: Any works relating to or affected by the discrepancies until the HP has been released.



## Status

- (d) The existing feature, level and services information provided on the drawings cannot be guaranteed. The Contractor must confirm the completeness and accuracy of all such information prior to commencement of works.

## Site survey

- (e) ROCKHAMPTON SPORTS PRECINCT – EXISTING CONDITIONS PLAN  
SURFCOAST Survey & Drafting Services P/L  
DECEMBER 2025 (labelled "MARCH 2025")  
2025-0110A ROCKHAMPTON SPORTS PRECINCT RevB

## Geotechnical investigation

- (f) Rockhampton Sports Precinct- Preliminary Site Investigation  
Construction Sciences  
27/10/2025  
P25-47 Revision 2  
Rockhampton Sports Precinct – Desktop PSI – MARev1
- (g) Rockhampton Sports Precinct Geotechnical Investigation  
CMW Geosciences  
31/03/2026 Revision 1  
NQL2024-0320 AC  
NQL2024-0320 AC Rev1 Rockhampton Sports Precinct

## 2.4 Capricorn Municipal Development Guidelines

### General

- (a) Capricorn Municipal Development Guidelines form the basis for the accepted standard of civil works, except where noted otherwise.
- (b) Where requirements differ between referenced documents, the more onerous specification/ requirement applies unless otherwise directed by the Superintendent.
- (c) Work shall adhere to the CMDG Construction specifications:

| No.             | Name                                 | Issue                       | Remarks                                      |
|-----------------|--------------------------------------|-----------------------------|----------------------------------------------|
| C201            | Control of Traffic                   | NO:3 – July 2018            |                                              |
| <del>C202</del> | <del>Rural Road Clear Zones</del>    | <del>NO:3 – July 2018</del> | Not used.                                    |
| C211            | Control of Erosion and Sedimentation | NO:2 – July 2018            |                                              |
| C212            | Clearing and Grubbing                | NO:3 – July 2018            |                                              |
| C213            | Earthworks                           | NO:3 – Aug 2024             | Refer Section 4 for additional requirements. |
| C220            | Stormwater Drainage                  | NO:3 – July 2018            | Refer Section 6 for additional requirements. |
| C221            | Pipe Drainage                        | NO:1 – October 2007         |                                              |
| C222            | Precast Box Culverts                 | NO:3 – July 2018            |                                              |

|                 |                                           |                              |                                                                                     |
|-----------------|-------------------------------------------|------------------------------|-------------------------------------------------------------------------------------|
| C223            | Drainage Structures                       | NO:3 – July 2018             | Refer Section 6 for additional requirements.                                        |
| C224            | Open Drains incl. Kerb & Gutter (Channel) | NO:1 – October 2007          | Refer Section 7 for additional requirements.                                        |
| C230            | Subsurface Drainage                       | NO:4 – August 2024           |                                                                                     |
| C231            | Subsoil and Foundation Drains             | NO:4 – July 2024             |                                                                                     |
| C232            | Pavement Drains                           | NO:2 – July 2018             |                                                                                     |
| <del>C233</del> | <del>Drainage Mats</del>                  | <del>NO:2 – July 2018</del>  | Not used.                                                                           |
| C241            | Stabilisation                             | NO:1 – October 2007          | Refer Section 5 for additional requirements and Annexure 241A.                      |
| C242            | Flexible Pavements                        | NO:1 – October 2007          | Refer Section 8 for additional requirements.                                        |
| C244            | Sprayed Bituminous Surfacing              | NO:1 – October 2007          | Refer Section 9 for additional requirements, and Annexures C244A, C244B, and C244C. |
| C245            | Asphaltic Concrete Surfacing              | NO:3 – Nov 2019              | Refer Section 9 for additional requirements and Annexure C245C.                     |
| C261            | Pavement Markings                         | NO:2 – July 2018             |                                                                                     |
| C262            | Signposting                               | NO:2 – July 2018             |                                                                                     |
| C263            | Guide Posts                               | NO:2 – April 2019            |                                                                                     |
| <del>C264</del> | <del>Guardrail</del>                      | <del>NO:2 – March 2019</del> | Not used.                                                                           |
| C265            | Boundary Fencing                          | NO:2 – July 2018             | Refer Section 11 for additional requirements.                                       |
| C271            | Minor Concrete Works                      | NO:2 – J 2018                | Refer Section 7 for additional requirements.                                        |
| C273            | Landscaping                               | NO:4 – Mar 2024              | Refer to project-specific Landscape specification.                                  |
| <del>C501</del> | <del>Bushfire Protection</del>            | <del>NO:3 – July 2018</del>  | Not used.                                                                           |

## 3. Specification Preliminaries

### 3.1 Conformance with Design

#### Quality management system

- (a) The Contractor shall develop and maintain a quality management system for all works under the Contract and shall comply with the requirements of AS/NZS ISO 9001.
- (b) Conformance of the works shall be demonstrated through systematic inspection and testing.
- (c) The requirements outlined in this section do not relieve the Contractor of the responsibility to conform with the Contract documents.
- (d) The Contractor is responsible for correcting all non-conformances.

#### Testing of materials and work

- (e) Compliance tests shall be carried out by the Contractor to ensure compliance with the specified requirements.
- (f) All compliance sampling and testing shall be carried out by laboratories accredited by NATA and certified for the appropriate tests, unless otherwise specified.
- (g) The frequency of testing for compliance shall not be less than the minimum specified.
- (h) Where a minimum testing frequency or minimum number of tests is not given, it shall be nominated by the Contractor and submitted to the Superintendent at least 14 days prior to the commencement of testing.
- (i) Test result submissions by the Contractor shall include a summary table listing all relevant specification requirements alongside the supplied test result(s), as well as noting whether the individual result is complying or non-complying.
- (j) Submissions that do not conform to the above will be rejected for re-submission.

#### Verification surveys

- (k) The Contractor shall undertake and submit verification surveys as specified in the Contract to demonstrate conformance with the drawings and specifications in relation to dimensions, tolerances and three-dimensional position.
- (l) Verification surveys shall be submitted with an accompanying survey conformance report where design levels, position and/or tolerances have been specified.
- (m) The report shall indicate the difference between actual and documented values for position and level (defined by co-ordinates or chainage and offset) and provide certification by the qualified surveyor responsible for the verification survey.

#### Non-conforming works

- (n) The Contractor is responsible for creating and maintaining a Non-Conformance Register.

- (o) Any works that depart from the documented requirements – whether identified by test result, verification survey, visual inspection or by any other means – shall be recorded by the Contractor on a Non-Conformance Report (NCR) form and on the Contractor's NCR register within two working days of detection / identification.
- (p) The NCR shall indicate the Contractor's proposed action, which may include:
  - i. Proposed additional works to bring the non-conforming works up to the documented standard.
  - ii. Proposed replacement of all or part of the non-conforming works to bring it up to the documented standard.
  - iii. For incidental defects, a request that the Superintendent accept the non-conforming works without alteration, as an exception with or without alteration to the respective contract sum.

**MILESTONE:**

- (q) At Practical Completion, or at any other time if requested by the Principal, provide a controlled copy of the Non-Conformance Register.

## 3.2 Quality Control Points

### Hold Point

- (a) A Hold Point is a point in the construction process beyond which the Contractor shall not proceed without written authorisation from the Superintendent.
- (b) Works to be held (i.e. that which the Contractor shall not proceed with until the Hold Point is released) are noted within each Hold Point. Where nothing is specified, all works are to be held.
- (c) The Contractor shall provide evidence to the Superintendent that all applicable work has been completed, tested, and inspected in accordance with the Contract.
- (d) The Superintendent's authorisation to proceed beyond the Hold Point does not relieve the Contractor of responsibility for satisfactory execution or performance of the work.

### Witness Point

- (e) A Witness Point is a point in the construction process where the Contractor must provide prior notice to the Superintendent, who may exercise the option of attendance to witness those works.
- (f) Unless specified otherwise, the Contractor shall give the Superintendent notice of at least 48 hours of an approaching Witness Point. The notice period may only commence during working hours.
- (g) The Contractor may proceed with the activity when the period of notice has expired whether or not the Superintendent elects to witness the activity.

- (h) The witnessing of an activity by the Superintendent or any delegate does not constitute approval or acceptance of the works.

### **Milestone**

- (i) A Milestone is a point in the construction process where progress is verified by the submission of information.

## **3.3 Site Establishment**

### **Site inception meeting**

#### **HOLD POINT AND WITNESS POINT**

- (a) Give notice of the proposed date for possession of site so that the Superintendent or their agent(s) may be in attendance for a site inception meeting.

HOLD: All works until HP has been released.

### **Existing infrastructure**

- (b) Do not obstruct or damage roadways and footpaths, drains and watercourses and other existing services in use on or adjacent to the site.

### **Site boundaries**

- (c) Site boundaries shall be delineated according to circumstances and the nature of work being carried out. Such delineation may include fences, barricades, warning signs and/or lights, locked doors/gates, witches' hats, total enclosure of the work site or other agreed methods.
- (d) Work areas, which are designated as restricted, shall display notices in accordance with the applicable Australian Standards for warning signs.
- (e) Email to be provided to SportEng with proof of signage.

### **Control survey**

#### **HOLD POINT**

- (f) Submit plan and details of the site control survey to be used for setting out.

HOLD: Site clearing and earthworks until HP has been released.

- (g) The control survey shall align with the height datum and co-ordinate system on the supplied site feature and level survey and documented on the drawings.
- (h) Verify all survey control marks in case of disturbance since installation.

- (i) If survey control is not provided by the Principal, conduct control survey in accordance with the recommended practices specified in Standard for the Australian Survey Control Network – SP1 Ver. 2.2 and its relevant guidelines ([www.icsm.gov.au/](http://www.icsm.gov.au/)).

### Setting out the works

- (j) Works shall be set out by a licensed surveyor to the design setout data / digital model.
- (k) Notify the Superintendent of any omission or conflict.
- (l) Layouts of service lines, plant and equipment as shown on the drawings are schematic only, except where figured dimensions or setout points are provided.

## 3.4 Operations and Maintenance Manuals

### General

#### MILESTONE

- (a) Prior to Practical Completion, provide final Operation and Maintenance Manuals in accordance with the requirements of this section.

### Objectives

- (b) The objectives of the Operations and Maintenance Manuals are to:
  - i. Be of sufficient detail to enable the Superintendent to take over any maintenance, operation or use of the works and to do so in a safe, effective and efficient manner
  - ii. Enable progressive and timely development and checking of the Manuals in advance of any completion milestones
  - iii. Be fully completed and finalised prior to the Superintendent's occupation, use or acceptance of the works
  - iv. Be developed in standardised and fully electronic data format suitable for upload to the Superintendent's Asset and Data Management Systems
  - v. Enable complete financial reconciliation of the assets and works showing element and asset costs, life expectancy costs and the like.

### Standard headings

- (c) Operations and Maintenance Manuals are to follow the standard headings shown below to ensure consistency for all elements of the works:
  - i. Introduction & Scope - description of the systems, the approach taken and other relevant information to ensure maintenance staff understand the equipment and its intended purpose
  - ii. Assets - detailed schedule of all maintainable assets data, items, and locations

- iii. Maintenance - detailed instructions and frequency to ensure proper function of the assets
- iv. Operations Data - detailed instructions for safe and efficient operation of the equipment
- v. Spare Parts - listed items or components required to complete maintenance or operation tasks or for replacements
- vi. Warranty and Certificates - descriptions of all warranties (both contracted and procured through suppliers) for the assets and descriptions of any certificates issued as part of the works including attached copies of all relevant documents
- vii. Help and Contact - Details of any relevant contractors, suppliers and the like who may be used by the owner to support the operation and maintenance of the assets
- viii. Drawings and Reference - lists of all final as-built drawings, specifications and other relevant documents forming the final contract scope and other relevant attachments - like product manuals, specifications and the like relevant to the proper operation and maintenance of the works.

(d) Where a particular section is not relevant it may be left blank.

### **Draft submission**

- (e) Collate all relevant information into a single report under the Standard Headings listed in this section.

#### **MILESTONE:**

- (f) Complete the draft Operations and Maintenance Manuals 28 days prior to proposed Practical Completion. Supply a draft for review by the Superintendent. The Superintendent may review Operation and Maintenance Manual and provide comments or directions for any corrections as needed.

- (g) Update the Operations and Maintenance Manuals for final submission prior to Practical Completion incorporating all comments or directions issued from the draft submission.

### **Compliance with laws, standards and specifications**

- (h) Check and verify that all data and attached files and documents that form the completed Operations and Maintenance Manuals comply with the relevant Laws, Standards, Codes and Specifications applicable to the works to enable the proper operation and maintenance by the Superintendent and / or its appointed agents of the completed works.

## 3.5 Completion

### Project signage

- (a) Install permanent project funding acknowledgement signage supplied by the Principal.
- (b) Installation locations:
  - i. At or near the Norman Road main entrance, as directed by the Principle.
- (c) Signage details:
  - i. 1 x 2000 x 1000 mm permanent external sign
  - ii. 1 x 1000 x 500 mm permanent internal sign.

#### MILESTONE

- (d) Provide details of completed signage installation with photos.

### As-constructed documentation

#### MILESTONE

- (e) Provide work-as-executed / as-constructed drawings in PDF and CAD formats showing the completed works as constructed and in accordance with the Specification requirements.
- (f) As-constructed documentation shall be produced from a survey of the completed works by a Licenced Surveyor using the Map Grid of Australia and Australian Height Datum.
- (g) As-constructed survey pick up is required for the line and layout of:
  - i. in-ground services, including irrigation, stormwater drainage
  - ii. pavements, kerbs, and edging
  - iii. fencing and sports equipment
  - iv. any other fixed elements or equipment.
- (h) Ensure the content, accuracy and level of detail of work-as-executed drawings are equivalent to those in the detail design drawings used for construction and are sufficient to describe the works adequate to enable future modifications or additions.
- (i) Where contract drawings are used as a base for production of the as-constructed documentation, all as-constructed details are to be clearly distinguishable from base / background information, e.g. via use of distinct colours, etc.
- (j) Any works backfilled without first being picked up by survey shall be uncovered to expose the top of the service, surveyed, and backfilled in accordance with the specification and the surface reinstated to the required condition.



**Product warranty submissions**

- (k) Where applicable, complete all documentation required by the manufacturer and submit those to the manufacturer to obtain the warranty and/or guarantee certificate(s).

**Defects liability period**

- (l) During the Defects Liability Period as defined in the Contract, the Contractor shall promptly rectify defects or omissions in the Contractor's works and are due to causes for which the Contractor is responsible.
- (m) The Contractor is not responsible for the repair, replacement or making good any defect or any damage to the works arising out of or resulting from any of the following causes:
  - i. Improper operation or maintenance of the works
  - ii. Use and operation of the works outside the Contract specifications.

## 4. Earthworks

### 4.1 General

#### Scope

- (a) This section sets out additional requirements for earthworks over and above relevant CMDG construction specifications, refer Section 2.4.

### 4.2 Supervision

#### Level 1 Supervision and Testing

- (b) The Contractor is required to engage a Geotechnical Inspection and Testing Authority (GITA) to perform Level 1 Inspection and Testing of all earthworks activities, as defined by AS 3798, Clause 8.2.
- (c) The Contractor's GITA must always have competent personnel on site while earthworks are undertaken, including (where applicable):
- i. Topsoil stripping
  - ii. Excavation and embankment construction
  - iii. Stabilisation (refer Section 5 Stabilisation)
  - iv. Moisture conditioning and compaction
  - v. Proof rolling.

#### HOLD POINT

- (d) On completion of the earthworks, provide a report authored by the Contractor's GITA setting out the inspections, sampling and testing it has carried out, and the locations and results thereof. The Contractor's GITA must express an opinion as to whether the works (as far as it has been able to determine) comply with the specification and drawings.

HOLD: All site works until HP has been released.

## 4.3 Materials

### Material types

(a) General fill:

|                              | Requirement |
|------------------------------|-------------|
| <b>Source</b>                | Site-won    |
| <b>Maximum particle size</b> | 75 mm       |

(b) Select fill:

|                                 | Requirement                  |
|---------------------------------|------------------------------|
| <b>Source</b>                   | Imported                     |
| <b>Grading</b>                  | Well-graded                  |
| <b>Maximum particle size</b>    | 75 mm                        |
| <b>Fines content</b>            | ≤ 25%-passing 0.075 mm sieve |
| <b>California Bearing Ratio</b> | ≥ 2%                         |
| <b>Swell</b>                    | ≤ 2.5%                       |
| <b>Plasticity index</b>         | 2 – 25                       |

### Imported materials

HOLD POINT

(c) Imported materials must be sampled at-source and tested prior to delivery to site. Tests must be current (i.e. performed within 3 months of the date supplied).

HOLD: Ordering and delivery of imported material until HP has been released.

### Unsuitable materials

(d) The following materials are unsuitable as fill or for fill to be constructed upon:

- i. organic materials, such as topsoils, severely root-affected subsoils and peat
- ii. silts, or materials that have the deleterious engineering properties of silt
- iii. material that contains boulders, wood, metal, plastic, or other deleterious material, in sufficient proportions to affect the required performance of the fill
- iv. materials contaminated through past site usage
- v. materials containing substances that can be dissolved or leached out in the presence of moisture (e.g. gypsum), or which undergo volume change or loss of strength when disturbed and exposed to moisture (e.g. shales, sandstones, etc.)
- vi. other materials with properties that are unsuitable for the forming earthworks.

Material properties testing

(e) Materials shall be tested to demonstrate compliance according to the following:

| Material / Location        | Test parameter                                      | Minimum Test Frequency                                                  |
|----------------------------|-----------------------------------------------------|-------------------------------------------------------------------------|
| General Fill<br>(imported) | Unsuitable materials and maximum particle dimension | Visual inspection and assessment / measurement of larger rock particles |
| Select Fill<br>(imported)  | Soil classification tests (plasticity and grading)  | 1 test per 500 m <sup>3</sup>                                           |
|                            | California Bearing Ratio (CBR) and Swell            | 1 test per 500 m <sup>3</sup>                                           |

## 5. Stabilisation

### 5.1 General

#### Scope

- (a) This section sets out additional requirements for stabilisation over and above relevant CMDG construction specifications, refer Section 2.4.

### 5.2 Additional Requirements

- (a) Stabilisation must be undertaken by an AustStab ARRB Accredited contractor.
- (b) Stabilisation layer shall be extended min. 1500 mm beyond edge of pavement extent, including any flush kerbs and grated trench drains, for the following pavement types:
- i. *Pavement Type 3: Asphalt Acrylic.*

### 5.3 Annexure C241A

|                                                                                        |                              |
|----------------------------------------------------------------------------------------|------------------------------|
| (a) Type of Stabilising Agent:                                                         | <b>Cement and lime</b>       |
| (b) Nominal Percentage of Stabilising Agent by Mass:                                   | To be determined via trials. |
| (c) Depth of Compacted Layer to be Stabilised:                                         | <b>350 mm</b>                |
| (d) Nominated Field Working Period:                                                    | To be determined via trials. |
| (e) Nominated Target Unconfined Compressive Strength (UCS) (7 day accelerated curing): | n/a                          |
| (f) Nominated Target CBR Value (4-day soaked) for stabilised modified subgrade:        | <b>15%</b>                   |
| (g) Period for Contractor's Curing:                                                    | To be determined via trials. |
| (h) Nominated Granular Material(s):                                                    | n/a                          |
| (i) Source of Nominated Granular Material:                                             | n/a                          |

## 6. Stormwater Drainage

### 6.1 General

#### Scope

- (a) This section sets out additional requirements for stormwater drainage over and above relevant CMDG construction specifications, refer Section 2.4.
- (b) The scope of this section includes:
  - i. Proprietary drainage systems.

### 6.2 Additional Requirements

#### Proprietary Drainage Systems

- (a) Proprietary drainage systems include grated trench drain systems, drainage cells, tank systems, stormwater treatment devices and the like.
- (b) Proprietary drainage systems shall be installed as shown on the drawings and in accordance with the manufacturer's requirements. Where conflicts exist, the more onerous requirement will apply.
- (c) Specified systems and products may only be varied if approved in writing by the Superintendent, following review by the design engineer.

## 7. In-situ Concrete

### 7.1 General

#### Scope

- (a) This section sets out additional requirements for in-situ concrete works over and above relevant CMDG construction specifications, refer Section 2.4.

### 7.2 Additional Requirements

#### Tolerances

- (a) Concrete shall be constructed to the following tolerances from design:

| Element                                                                        | Description                       | Tolerance                                        |
|--------------------------------------------------------------------------------|-----------------------------------|--------------------------------------------------|
| <b>Kerbs and concrete forming an edge for Pavement Type 3: Asphalt Acrylic</b> | Horizontal position               | ±10 mm                                           |
|                                                                                | Surface level                     | ±5 mm                                            |
|                                                                                | Rate of change from line or level | No greater than 5 mm beneath a 3 m straight edge |

- (b) Notwithstanding application of construction tolerances generally, all pedestrian paths shall conform to the requirements of AS 1428.1 and AS 1428.2.

#### Underlay

- (c) Under slabs on ground, provide a vapour barrier underlay consisting of 200 µm (0.2 mm) thick polyethylene film, medium impact resistance, complying with the requirements of AS 2870 Section 5.3.3.
- (d) Underlay shall entirely cover the extent of slab and extend under formwork.
- (e) Lapping for continuity at joints shall be not less than 200 mm.

#### Appearance and surface finish

- (f) Concrete colouring and surface finishes to Landscape Architect's specifications.

#### Joints

- (g) Saw-cut (contraction) joints in concrete pavements shall be made within 12 – 24 hours of the concrete being placed.
- (h) Isolation joints shall be constructed at all interfaces with rigid structures such as buildings, pits, adjacent concrete elements, and the like.
- (i) Dowels used in expansion joints shall be galvanised.
- (j) All concrete expansion and isolation joints shall be sealed with polyurethane sealant in accordance with the manufacturer's requirements, unless noted otherwise.

## 8. Flexible Pavements

### 8.1 General

#### Scope

- (a) This section sets out additional requirements for flexible pavement works over and above relevant CMDG construction specifications, refer Section 2.4.
- (b) Additional requirements apply for the base and sub-base layers in the following pavement types:
  - i. *Pavement Type 3: Asphalt Acrylic.*

### 8.2 Additional Requirements

#### Tolerances

- (a) Base and sub-base layers for *Pavement Type 3: Asphalt Acrylic* shall be constructed to the following tolerances from design:

| Element  | Description   | Tolerance                                        |
|----------|---------------|--------------------------------------------------|
| Base     | Surface level | +0 / -10 mm                                      |
|          | Thickness     | +5 / -0 mm                                       |
|          | Flatness      | No greater than 5 mm beneath a 3 m straight edge |
| Sub-base | Surface level | ±5 mm                                            |
|          | Thickness     | +5 / -0 mm                                       |

#### Proof rolling

- (b) Prior to priming and / or placement of overlying material, proof roll base and sub-base layers for *Pavement Type 3: Asphalt Acrylic*.
- (c) Proof rolling shall be carried out in accordance with *CMDG Construction Specification Clause C213.30.07*.
- (d) Pavement layers shall be compacted to be capable of withstanding proof rolling without visible deformation or springing.

#### HOLD POINT AND WITNESS POINT

- (e) The Contractor's GITA shall witness all proof rolling required under this specification. The GITA shall supply a test report and / or site inspection report indicating whether the proof rolling identified any areas of deformation or springing to be rectified.

HOLD: Any works on or within layers to be proof rolled until HP has been released.



## Verification Survey

- (f) The surfaces of all pavement layers shall be surveyed to verify the completed works comply with the design and tolerances (refer *Section 8.2 Conformance with Design*).

## Methodology

- (g) Verification surveys shall comply with the general requirements of *Section 3.1 Conformance with Design*.
- (h) Pavement layer survey measurements to be taken:
- i. Along lines of kerb, edging, pavement edges, steps, ramps, grated trench drains, and the like, at spacing not greater than 3 metres.
  - ii. Pavement spot levels on a grid with spacing not greater than 3 x 3 metres
  - iii. At all changes in direction, changes in grade, steps, and other features.

### HOLD POINT

- (i) Submit a verification survey report comparing the as-built levels with design levels, including highlighting any points that exceed the tolerances (refer *Section 8.2(a)*).

HOLD: Priming, placement of overlying pavement layers, and construction of concrete kerbs and edging (where applicable) until HP has been released.

## 9. Asphalt and Bituminous Surfacing

### 9.1 General

#### Scope

- (a) This section sets out additional requirements for asphalt and bituminous surfacing works over and above relevant CMDG construction specifications, refer Section 2.4.
- (b) Additional requirements apply for the asphalt layer in the following pavement types:
  - i. *Pavement Type 3: Asphalt Acrylic.*

### 9.2 Additional Requirements

#### Tolerances

- (a) Asphalt acrylic pavement surfaces shall adhere to the following tolerances from design:

| Element                                                              | Description   | Tolerance                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pavement Type 3:<br/>Asphalt Acrylic<br/><br/>(asphalt layer)</b> | Surface level | ±5 mm                                                                                                                                                                                                                                                           |
|                                                                      | Thickness     | +5 / -0 mm                                                                                                                                                                                                                                                      |
|                                                                      | Flatness      | <p>No greater than 4 mm beneath a 3 m straight edge in all directions.</p> <p>No greater than 2 mm under a 1 m straight edge in all directions.</p> <p>No step like irregularities greater than 1 mm.</p> <p>Surface to be free draining (i.e. no ponding).</p> |

- (b) Notwithstanding application of construction tolerances generally, all pedestrian paths shall conform to the requirements of AS 1428.1 and AS 1428.2.

#### Aggregate and foreign materials

- (c) Aggregate used for asphalt in *Pavement Type 3: Asphalt Acrylic* shall exhibit negligible pyrite concentration and be free of any other foreign substances that could lead to staining, blister sports, or other defects in the acrylic surface system.

HOLD POINT:

- (d) Submit evidence from a local asphalt supplier that the geographical source of their product is known to be free of ferrous bearing contaminants (i.e. pyrites).

HOLD: Delivery and placement of asphalt for *Pavement Type 3* until HP has been released.

### Drying back pavement base

WITNESS POINT

- (e) After preparation of the surface and prior to priming, the uppermost 10 mm of granular material shall be allowed to dry back to a moisture content less than 60% of OMC.

### Visual inspection for asphalt stains

- (f) If staining of the surface is visible after asphalt has been laid – due to pyrites, or any other foreign materials – perform one or more of the following:

- i. Remove and replace the affected asphalt (minimum depth 25 mm)
- ii. Apply stain blocker across the entire asphalt surface.

HOLD POINT

- (g) Submit the extent of proposed remediation and methodology, prior to the commencement of works.

HOLD: Verification survey and flood test (if applicable) until HP has been released.

### Verification survey

- (h) Verification surveys shall comply with the general requirements of *Section 3.1 Conformance with Design*.
- (i) Pavement layer survey measurements to be taken:
  - i. Along lines pavement edges, and the like, at spacing not greater than 3 metres.
  - ii. Pavement spot levels on a grid with spacing not greater than 3 x 3 metres
  - iii. At all changes in direction, changes in grade, steps, and other features.

HOLD POINT

- (j) Submit a verification survey report comparing the as-built levels with design levels, highlighting any points that exceed the tolerances of Section 9.2.

HOLD: Works on or associated with the given asphalt layer until HP has been released.

## Flood test

### HOLD POINT AND WITNESS POINT

- (k) Following completion of the asphalt layer for *Pavement Type 3: Asphalt Acrylic*, and prior to any additional surfacing works, a flood test shall be undertaken.
- (l) Flood the entire asphalt surface with potable water by any appropriate means in the presence of the Superintendent.

HOLD: Works on or associated with the given asphalt layer until HP has been released.

- (m) Identify and document any defects visible on the flooded surface including the following:
  - i. depressions that may hold water
  - ii. high-points, ridges, or similar that may divert or prevent surface water run-off.

## Remediation works

- (n) Undertake remedial works if any defects are identified.

### HOLD POINT

- (o) Provide the proposed remediation methodology prior to works commencing.

HOLD: Remedial works until HP has been released.

- (p) Repeat the flood test and verification survey following remedial works.

## 9.3 Annexure C244A

- (a) Details of sprayed bituminous surfacing shall be as shown on the Contract drawings.

## 9.4 Annexure C244B

- (a) Binder details shall be as shown on the Contract drawings.
- (b) Primer, Primerbinder and Binder Application Tolerance Thresholds T2:
  - i. 0 mm prime:  $\pm 10\%$
  - ii. 10 mm primer seal:  $\pm 10\%$ .

## 9.5 Annexure C244C

- (a) Quality Control and Testing for sprayed bituminous surfacing shall include:
  - i. Audit testing: Required.

- |      |                                         |                             |
|------|-----------------------------------------|-----------------------------|
| ii.  | Audit testing – maximum lot size:       | 1 per Pavement Type.        |
| iii. | Audit testing – minimum test frequency: | 1 per 5000 m <sup>2</sup> . |

## 9.6 Annexure C245C

- (a) Nominal sizes of asphalt required for this contract (tick box) and enter binder type:
- |      |                                             |  |
|------|---------------------------------------------|--|
| i.   | AC7 (DG7) – Class 320 Binder                |  |
| ii.  | AC10 (AC10) – Class 320 Binder              |  |
| iii. | AC14 (AC14) – A15E Polymer Modified Binder. |  |
- (b) Nomination of aggregate pre-treatment procedure if required by Superintendent: TBC.
- (c) Special aggregate mixes required for this contract: Stain (e.g. pyrite)-free mix, refer clause 9.2(c) and (f).
- (d) Requirements for removal of thermoplastic or other line marking: None.
- (e) Requirement for Preparation of Surface Hold Point Refer Clause C245.21 PREPARATION OF PAVEMENT: Yes.

## 10. Acrylic Surfacing

### 10.1 General

#### Scope

- (a) This section sets out the technical and verification requirements for acrylic surfacing to be executed under the Contract.
- (b) The scope of this section includes:
  - i. Installation, testing, and certification of acrylic surfacing system to meet all requirements of the relevant sports governing body standards.

#### Standards

- (c) Works shall meet all quality and performance requirements of the following documents:
  - i. Netball Australia, National Facilities Policy, Version 02 March 2016
  - ii. International Netball Federation (INF), Facilities and Equipment Requirements, July 2015
  - iii. AS 4586—2013 Slip Resistance Classifications of New Pedestrian Surface Materials
  - iv. AS 4663—2013 Slip Resistance Classification of Existing Pedestrian Surface Materials
  - v. FIFA Futsal Laws of the Game, current edition.

#### Project particulars

- (d) Surface type: liquid-applied acrylic surface system.
- (e) Colour scheme:
  - i. Netball court surface: green, red and blue as shown on drawings
  - ii. Netball line marking: white
  - iii. Pickleball court surface: green and blue as shown on drawings
  - iv. Pickleball line marking: orange
  - v. Multi-use court surface: green
  - vi. Multi-use line marking: white.
- (f) Lines and marks shall be repainted at the end of the Defects Liability Period.
- (g) The Contractor shall allow for the surface to be re-tested 3 years after installation:

- i. testing shall be in accordance with Section 10.5
- ii. any failure to achieve the performance requirements shall be considered a construction defect and must be rectified at the cost to the Contractor
- iii. satisfactory rectification shall be demonstrated by submission of a complete set of conforming test results and certificate (repeat testing may be necessary).

## 10.2 Conformance with Design

### Quality Management System

- (a) The Contractor shall implement a quality management system for all works under the Contract in accordance with *Section 3 Specification Preliminaries*.

### Acceptance criteria

- (b) Acceptance of the acrylic surface is subject to:
  - i. visual inspection, exhibiting a high quality of workmanship
  - ii. performance testing (refer *Section 10.5*)
  - iii. verification survey (refer *Section 10.6*)
  - iv. flood test (refer *Section 10.7*).

### Tolerances

- (c) Acrylic surfaces shall adhere to the following tolerances from design:
  - i. Surface level:  $\pm 5$  mm
  - ii. Flatness: No greater than 4 mm beneath a 3 m straight edge in all directions.  
  
No greater than 2 mm under a 1 m straight edge in all directions.  
  
No step like irregularities greater than 1 mm.  
  
Surface to be free draining (i.e. no ponding from flood test, refer to *Section 10.7 Flood Test*).
- (d) Line marking in the acrylic surface shall adhere to the following tolerances from design:
  - i. Straightness:  $\pm 5$  mm over any 10m length  
No horizontal step-like irregularities greater than 0.5 mm
  - ii. Location:  $\pm 5$  mm

## 10.3 Materials

### General

#### HOLD POINT

- (a) Provide technical details, including product technical and safety data sheets, for all elements of the proposed acrylic court system.
- (b) Provide physical samples for the nominated colours, 100 x 100 mm minimum size.

HOLD: Installation of acrylic products until HP has been released.

### Acrylic surface system

- (c) The surface system shall be:
  - i. suitable for the climatic conditions and environment of the proposed site
  - ii. reduce surface temperature through enhanced solar reflection (verified via independent testing)
  - iii. able to comply with all sports governing body requirements for outdoor tennis courts for all categories of competition and training
  - iv. able to sustain regular and intensive use
  - v. selected to avoid delamination from either the base strata (asphalt/concrete) or between acrylic layers
  - vi. consist of a stain blocking primer layer to ensure no staining from the underlying impurities in the asphalt layer.
- (d) All products shall be delivered to site in sealed, clearly identified containers.
- (e) Containers shall be stored in a dry, shaded location (e.g. site shed, under awning, etc.) not subject to rainfall.
- (f) The colour scheme for the acrylic surface shall be as specified in the project particulars, refer *Section 10.1*.

## 10.4 Installation

### General

#### HOLD POINT

- (a) Provide layout plan showing dimensions and colours of all areas, lines, and marks.

HOLD: Preparation and application of acrylic surface until HP has been released.



## Surface application

- (b) The installation methods shall ensure the resulting surface is of the highest standard and quality possible.
- (c) Repair any cracks in the base (if required), provide methodology for crack repair for approval prior to commencement.
- (d) Prepare the surface by cleaning, filling, and priming as required.
- (e) Throughout installation, ensure the base surface remains clean and clear of debris.
- (f) Apply products in accordance with the product manufacturer's recommendations and requirements to ensure uniformity and quality.
- (g) Do not apply in adverse weather conditions that may affect the installation quality.

### WITNESS POINT

- (h) Each layer of the surface shall be applied to a high standard and be free of streaks, patches, and visible squeegee applicator directions.

## Lines and marks

- (i) Lines and marks shall be in accordance with the drawings and relevant sports governing body standards.
- (j) Set out of the lines shall be undertaken by a licensed surveyor.
- (k) All lines and marks shall be:
  - i. 50 mm wide, unbroken lines
  - ii. coloured as specified in the project particulars, refer *Section 10.1 General*
  - iii. painted with approved acrylic material compatible with the acrylic surface
  - iv. applied according to the manufacturer's recommendations, flush with the surface of the court, with distinct edges and without overspray or bleeding of the paint.

## 10.5 Performance Testing

### General

- (a) The completed court surface must be fit for purpose, which includes 'all weather' use in outdoor settings.
- (b) All testing shall be undertaken by a NATA accredited testing company.

### Slip resistance for FIFA compliance

- (c) A minimum of 5 individual locations on each court shall be tested for slip-resistance.

- (d) The acrylic surface shall meet the following performance standard:
- i. A temperature correction value (TCV) for a mean British Pendulum Number (BPN) of at least 60 for wet slip resistance testing - in line with AS 4663-2013 Slip Resistance measurement of existing pedestrian surfaces.

### **Slip resistance testing for INF compliance**

- (e) A minimum of 5 individual locations on each court shall be tested for slip-resistance:
- i. wet and dry conditions
  - ii. slider 55 and slider 96
  - iii. making a total of 4 tests in each location.
- (f) The acrylic surface shall meet one of the following performance standards:
- i. A temperature correction value (TCV) for a mean British Pendulum Number (BPN) of at least 75 for wet slip resistance testing - in line with AS 4663-2013 Slip Resistance measurement of existing pedestrian surfaces.
  - ii. AS 4586-2013 Slip Resistance Classification of New Pedestrian Surface Materials: Appendix A (Wet Pendulum Test Method) achieving a P5 Classification as a minimum.

#### **MILESTONE**

- (g) Provide slip resistance test results demonstrating compliance with the performance requirements.

## 10.6 Verification Survey

### Extent

- (a) The completed acrylic surface(s) shall be surveyed to verify the completed works comply with the design and tolerances (refer *Section 10.2 Conformance with Design*).

### Methodology

- (b) Surface survey measurements to be taken:
  - i. At the start, end, and at maximum 3 metre intervals along all lines and marks
  - ii. At all changes in grade, edges of pavement, and other features
  - iii. Spot levels on a grid with spacing not greater than 3 x 3 metres.
- (c) The survey shall incorporate any additional measurements required to demonstrate compliance with the relevant sports governing body standards.

#### MILESTONE

- (d) Submit a verification survey report comparing the as-built levels and grades with design, including highlighting any points that exceed the tolerances (refer *Section 10.2 Conformance with Design*).
- (e) Provide the survey as an as-built plan of linemarking and acrylic court layouts.

## 10.7 Flood Test

### General

#### MILESTONE

- (a) A flood test shall be undertaken at completion of the acrylic surfacing works.

### Methodology

#### WITNESS POINT

- (b) Flood the entire surface with potable water by any appropriate means.
- (c) Identify and document any defects visible on the flooded surface including the following:
  - i. Depressions that may hold water

- ii. High-points, ridges or similar that may divert and / or prevent surface water run-off.

### Remedial works

- (d) Undertake remedial works if any of the defects are identified.

#### HOLD POINT

- (e) Provide the proposed remediation methodology prior to works commencing.

HOLD: Remedial works until HP has been released.

- (f) Repeat the flood test following remedial works.

## 10.8 Maintenance Manuals and Warranty

### Maintenance Manuals

#### MILESTONE

- (a) Provide electronic (PDF format) maintenance manuals for the acrylic surface(s) installed under the Contract prior to Practical Completion.

- (b) Maintenance manuals must specify:
  - i. all routine and periodic maintenance procedures
  - ii. limitations of use / intended uses for preservation of warranty.

### Instructional session

#### WITNESS POINT

- (c) Provide a single instructional session for the acrylic court system attended by up to two people as nominated by the Superintendent, to outline and demonstrate key elements of the maintenance manuals.

### Warranty

#### MILESTONE

- (d) The Contractor shall provide a 7-year warranty in favour of the Principal.

- (e) The following shall be warranted:

- i. The acrylic surface shall remain compliant with the tolerances and performance requirements of the nominated sports governing body throughout the warranty period.

## **11. Fencing, Netting, and Enclosures**

### **11.1 General**

#### **Scope**

- (a) This section sets out the technical and verification requirements for fencing, netting, and chain mesh enclosure works to be executed under the Contract.
- (b) The scope of this section includes:
  - i. installation of fencing, enclosures, gates, and all associated equipment
  - ii. detailed design of 1.05m, 1.8m and 2.4m High Chain Link Fence (including footings) in accordance with the reference design drawings(s) and specified performance requirements
- (c) Where works are within the scope of the National Construction Code (NCC), all permits and certifications in accordance with local laws and regulations is the responsibility of the Contractor.

#### **Referenced standards**

- (d) AS 1627.4–1989 Metal finishing - Preparation and pre-treatment of surfaces
- (e) AS 1725.1–2010 Security fences and gates – General requirements
- (f) AS 1725.2–2010 Tennis court Fencing – Commercial
- (g) AS 1725.4–2010 Chain link fabric fencing – Cricket net fencing enclosures
- (h) AS 1725.5–2010 Sports ground fencing – General requirements
- (i) AS 4100–2020 Steel structures
- (j) AS 4506–2005 Metal finishing – Thermoset powder coatings

## 11.2 Conformance with Design

### Quality Management System

- (a) The Contractor shall implement a quality management system for all works under the Contract in accordance with *Section 3 Specification Preliminaries*.

### Tolerances

- (b) Works shall be constructed to the following tolerances from design:

| Element                  | Description                                        | Tolerance                                                 |
|--------------------------|----------------------------------------------------|-----------------------------------------------------------|
| <b>Posts</b>             | Horizontal position                                | $\pm 20$ mm                                               |
|                          | Height                                             | $\pm 10$ mm                                               |
|                          | Straightness                                       | $\leq 2$ mm per metre from true vertical                  |
| <b>Rails</b>             | Vertical position                                  | $\pm 50$ mm                                               |
|                          | Straightness                                       | $\leq 3$ mm per metre from true horizontal                |
| <b>Suspended netting</b> | Support height                                     | $\pm 20$ mm                                               |
| <b>Tension Cables</b>    | Sag                                                | $\leq 100$ mm from a straight line between cable supports |
| <b>Clearance</b>         | Fence to ground-level and / or adjacent structures | 20 – 40 mm, unless noted otherwise on the drawings        |

- (c) Notwithstanding the above tolerances, no part of the fencing, netting, enclosure, or associated equipment shall encroach into the minimum specified unobstructed areas surrounding fields of play (i.e. run-off zones).

## 11.3 Materials

### General

- (a) All materials to comply with the referenced Australian Standard(s).
- (b) Fencing, netting, enclosures, and associated equipment shall have a functional life of:
  - i. 10 years for chain link fabric (including tie wires)
  - ii. 15 years for netting fabric and suspension cables
  - iii. 20 years for structural components (footings, posts, rails, brackets, and fixings).

### Footings

- (c) All concrete post footings shall be normal class (N class) with a standard strength grade of 25 MPa (28-day compressive strength) and maximum aggregate size of 20 mm.

### Steel

- (d) All steel components shall comply with AS 4100 Steel Structures.
- (e) All fittings and fixtures shall be galvanised or stainless steel.

### Posts and Rails

- (f) All posts, rails and associated fittings shall be black powder-coated galvanised steel, or otherwise shall be stainless steel.
- (g) All elements must be powder coated (or painted if specified) prior to installation.

### Chain Link Fabric

- (h) Chain link fabric to be heavy duty fabric made from 3.15mm black fuse-bonded polymer-coated wire
- (i) Chain link fabric pitch shall be:
  - i. 40 mm for the rear of all cricket practice net enclosures
  - ii. 45 mm for facilities containing tennis court(s)
  - iii. 50 mm pitch for all other installations.

### Painted Elements

- (j) Elements noted as 'painted' shall be prepared as follows:
  - i. Surface preparation: Class 2.5 blast to AS 1627.4.
  - ii. Single coat Dulux: Durebuild STE to a minimum dry thickness of 125 micron.
  - iii. Single coat Dulux: Weathermax HBR to a minimum dry thickness of 100 micron.



## 11.4 Installation

### General

- (a) Construct fencing, netting, enclosures, and associated equipment in locations and for the extent shown on the drawings.
- (b) Construct pedestrian and vehicle access gates in locations shown on the drawings.

#### WITNESS POINT

- (c) All installations to comply with the referenced Australian Standard(s), the drawings and this specification.

## 11.5 Maintenance Manuals and Warranty

### Maintenance Manuals

#### MILESTONE

- (a) Provide electronic (PDF format) maintenance manuals for all installed fencing, netting, enclosures, and associated equipment prior to Practical Completion.
- (b) Maintenance manuals must specify:
  - i. all routine and periodic maintenance procedures
  - ii. limitations of use / intended uses for preservation of warranty.

### Warranty

#### MILESTONE

- (c) The Contractor shall provide documentation of all necessary warranties prior to Practical Completion.
- (d) All fencing, netting, enclosures, and associated equipment shall be warranted in favour of the Principal for the following minimum timeframes:
  - i. fencing, netting, enclosures, and associated equipment: 5 years
- (e) The following shall be warranted:
  - i. performance of the equipment to satisfy the Principal's requirements
  - ii. any faults due to poor workmanship in the manufacture and installation of the equipment.

## **12. Sports Equipment – Netball**

### **12.1 General**

#### **Scope**

- (a) This section sets out the technical and verification requirements for Sports Equipment – Netball works to be executed under the Contract.
- (b) The scope of this section includes:
  - i. Supply and installation of sports equipment as nominated on the drawings.
  - ii. Provision of maintenance manuals and warranty documentation for the sports equipment.

#### **Standards**

- (c) International Netball Federation (INF) Facilities and Equipment Requirement.

#### **Quality Management System**

- (d) The Contractor shall implement a quality management system for all works under the Contract in accordance with *Section 3 Specification Preliminaries*.

#### **Tolerances**

- (e) All installed products must meet the comply with and/or pass the relevant test methods issued by the sport governing body.

## 12.2 Products and Materials

### General

#### HOLD POINT

- (a) Provide product details including manufacturer, product name/number, and samples if required, for each element to be installed under the contract.

HOLD: Ordering and installation of sports equipment until HP has been released.

### Netball Goals

- (b) Netball goals shall be semi-permanent with sleeved footings.
- (c) Posts shall be powder-coated white, galvanised steel with outside dia. 60 – 100 mm.
- (d) Netball goals shall be 3.05 metres (finished court surface to rim of goal).
- (e) Semi-permanent post bases (where specified) shall be set into concrete footings in accordance with the manufacturer's requirements.
- (f) Supply removable covers for post bases that provide a fit for purpose surface when goals are not installed.
- (g) Safety padding shall be supplied with all goal and point posts:
  - i. 3.0 m min. height, 50 mm thickness.
- (h) Example products meeting the above requirements (in alphabetical order):
  - i. Abel Sports PPF-NB3000/ PPC-NB3000 Netball Post Pads
  - ii. Play Hard Sports NS60 Senior height – 60mm dia. post installed with EHC hinged ground sleeves.

## 12.3 Installation

### General

- (a) All equipment shall be installed in accordance with the manufacturer's requirements.
- (b) Site-specific design of structural footings is the responsibility of the Contractor.
- (c) Where works are within the scope of the National Construction Code (i.e. Class 10b structures), it is the responsibility of the Contractor to obtain permits and certifications in accordance with local laws and regulations.

HOLD POINT

- (d) Provide shop drawings and building permit documentation (if required) prior to ordering and installation of equipment.

HOLD: Ordering of products and materials until HP has been released.

## 12.4 Maintenance Manuals and Warranty

### Maintenance Manuals

MILESTONE

- (a) Provide electronic (PDF format) maintenance manuals for all sports equipment supplied and/or installed under the Contract prior to Practical Completion.

- (b) Maintenance manuals must specify:

- i. all routine and periodic maintenance procedures
- ii. semi-permanent post removal and reinstatement methodology, including fit-for-purpose surface reinstatement when goal posts and down and field-of-play is reconfigured for other sporting activities
- iii. limitations of use / intended uses for preservation of warranty.

WITNESS POINT

- (c) Co-ordinate a training session with Principal's agents demonstrating semi-permanent post removal and reinstatement methodology, including fit-for-purpose surface reinstatement when goal posts and down and field-of-play is reconfigured for other sporting activities.

### Warranty

MILESTONE

- (d) The Contractor shall provide a minimum 10-year warranty in favour of the Principal.

- (e) The following shall be warranted:

- i. Equipment shall remain fit for the intended purpose and comply with the relevant sport governing body requirements (current at time of installation) for the duration of the warranty period.
- ii. Faults due to poor workmanship in the manufacture and / or installation of the equipment.

## 13. Sports Equipment – Pickleball

### 13.1 General

#### Scope

- (a) This section sets out the technical and verification requirements for Sports Equipment - Pickleball to be supplied and installed under the Contract.
- (b) The scope of this section includes:
  - i. Supply and installation of sports equipment as nominated on the drawings.
  - ii. Provision of maintenance manuals and warranty documentation for the sports equipment.

#### Standards

- (c) Pickleball Australia Facility Guidelines – 2025 Version 1 – updated March 2026.

#### Quality Management System

- (d) The Contractor shall implement a quality management system for all works under the Contract in accordance with *Section 3 Specification Preliminaries*.

#### Tolerances

- (e) All installed products must meet the comply with and/or pass the relevant test methods issued by the sport governing body.

### 13.2 Products and Materials

#### General

##### HOLD POINT

- (a) Prior to ordering any equipment, provide the relevant specifications of each element, including samples for colour selection.

HOLD: Ordering and installation of sports equipment until HP has been released.

#### Pickleball Posts and Nets

- (b) Permanent net posts, complete with:
  - i. 76 x 76 x 2.6 SHS galvanised net posts
  - ii. powder coated (black colour)
  - iii. internal winder mechanism with handle, stainless steel fittings

- iv. baseplate or other permanent fixing mechanism.
- (c) Standard nets, complete with:
  - i. length 6.7 m
  - ii. height 0.914 m at sidelines, 0.864 m centre
  - iii. 3.5mm resin treated & UV stabilised braided twine
  - iv. 6 mm (8 mm max.) polymer coated steel headline wire
  - v. white polyester canvas and polyester stitched headband 50 - 63.5 mm deep.
- (d) Example manufacturers known to produce products that meet the above specifications (in alphabetical order):
  - i. Play Hard Sports.

## 13.3 Installation

### General

- (a) All products shall be installed in accordance with the manufacturer's requirements.

#### HOLD POINT

- (b) Contractor to provide shop drawings prior to delivery of products and materials.

HOLD: Ordering of products and materials until HP has been released.

- (c) Site-specific design of structural footings is the responsibility of the Contractor.
- (d) Where works are within the scope of the National Construction Code (i.e. Class 10b structures), it is the responsibility of the Contractor to obtain permits and certifications in accordance with local laws and regulations.

## 13.4 Maintenance Manuals and Warranty

### Maintenance Manuals

#### MILESTONE

- (a) Provide electronic (PDF format) maintenance manuals for all sports equipment supplied and/or installed under the Contract prior to Practical Completion.

- (b) Maintenance manuals must specify:
  - i. all routine and periodic maintenance procedures

- ii. semi-permanent post removal and reinstatement methodology, including fit-for-purpose surface reinstatement when goal posts and down and field-of-play is reconfigured for other sporting activities
- iii. limitations of use / intended uses for preservation of warranty.

WITNESS POINT

- (c) Co-ordinate a training session with Principal's agents demonstrating semi-permanent post removal and reinstatement methodology, including fit-for-purpose surface reinstatement when goal posts and down and field-of-play is reconfigured for other sporting activities.

**Warranty**

MILESTONE

- (d) The Contractor shall provide a minimum 10-year warranty in favour of the Principal.

- (e) The following shall be warranted:

- i. Equipment shall remain fit for the intended purpose and comply with the relevant sport governing body requirements (current at time of installation) for the duration of the warranty period.
- ii. Faults due to poor workmanship in the manufacture and / or installation of the equipment.

## 14. Sports Equipment – Multi-Use Goals

### 14.1 General

#### Scope

- (a) This section sets out the technical and verification requirements for Sports Equipment – Combined Basketball and Futsal works to be executed under the Contract.
- (b) The scope of this section includes:
  - i. detailed design of combined basketball and futsal goals in accordance with the reference design drawings(s) and specified performance requirements
  - ii. Supply and installation of sports equipment as nominated on the drawings.
  - iii. Provision of maintenance manuals and warranty documentation for the sports equipment.

#### Standards

- (c) AS 1627.4–1989 Metal finishing - Preparation and pre-treatment of surfaces
- (d) AS 4100–2020 Steel structures
- (e) AS 4506–2005 Metal finishing – Thermoset powder coatings
- (f) AS 4685.1–2014 Playground equipment and surfacing – Part 1: General safety requirements and test methods
- (g) FIBA 2022 Official Basketball Rules – Basketball Equipment
- (h) FIFA Futsal Laws of the Game 2024-25.

#### Quality Management System

- (i) The Contractor shall implement a quality management system for all works under the Contract in accordance with *Section 3 Specification Preliminaries*.

#### Tolerances

- (j) All installed products must meet the comply with and/or pass the relevant test methods issued by the sport governing body.

#### Design & Construct Performance Requirements

- (k) Footings shall be designed to:
  - i. suit the site-specific geotechnical conditions (Contractor to allow for the cost of any necessary additional geotechnical investigations for this purpose)
  - ii. provide minimum mandatory clearances to existing and proposed in-ground services and drainage (150 mm minimum clearance if not otherwise stipulated)



- iii. not encroach into the field of play profile and be set below FSL.
- (l) The combined goals shall be designed and installed as a complete system fit for the intended purpose as documented, and comply with the following:
  - i. functional life of 20 years
  - ii. permanent installations
  - iii. black powder-coated galvanised steel (unless noted otherwise)
- (m) All ground-level posts shall be encased in a concrete slab or concrete edge strip of suitable width to surround the full width of the post (and fixing system, if applicable).

### Design Documentation

#### HOLD POINT

- (n) Submit all design and building permit documentation, including shop drawings, engineering calculations, building permit, and any associated specifications for the system.

HOLD: Ordering and installation of goals until the HP has been released.

## 14.2 Products and Materials

### General

#### HOLD POINT

- (a) Provide product details including manufacturer, product name/number, and samples if required, for each element to be installed under the contract.

HOLD: Ordering and installation of sports equipment until HP has been released.

### General

- (b) All materials to comply with the referenced Australian Standard(s).

### Footings

- (c) All concrete post footings shall be normal class (N class) with a standard strength grade of 25 MPa (28-day compressive strength) and maximum aggregate size of 20 mm.

### Steel

- (d) All steel components shall comply with AS 4100 Steel Structures.
- (e) All elements, fittings, and fixtures shall be galvanised or stainless steel.

## Combined Basketball and Futsal goals

- (f) Combined basketball and futsal goals shall be supplied as specified on the drawings.
- (g) The combined goal units shall comprise of a basketball tower positioned above a futsal goal with the following features:
- (h) Basketball tower component:
  - i. Fixed 3.05m high ring with backboard
  - ii. 1.20m outreach into the court
  - iii. 300kg (min.) load rated ring
- (i) Futsal goal component:
  - i. 3.0 m wide x 2.0 m high opening
  - ii. Square backbar on ground
  - iii. Fully enclosed by rigid steel mesh (25 mm opening).

## 14.3 Installation

### General

- (a) All equipment shall be installed in accordance with the manufacturer's requirements.
- (b) Site-specific design of structural footings is the responsibility of the Contractor.
- (c) Where works are within the scope of the National Construction Code (i.e. Class 10b structures), it is the responsibility of the Contractor to obtain permits and certifications in accordance with local laws and regulations.

#### HOLD POINT

- (d) Provide shop drawings and building permit documentation (if required) prior to ordering and installation of equipment.

HOLD: Ordering of products and materials until HP has been released.

## 14.4 Maintenance Manuals and Warranty

### Maintenance Manuals

#### MILESTONE

- (a) Provide electronic (PDF format) maintenance manuals for all sports equipment supplied and/or installed under the Contract prior to Practical Completion.

- (b) Maintenance manuals must specify:
- i. all routine and periodic maintenance procedures
  - ii. limitations of use / intended uses for preservation of warranty.

## Warranty

### MILESTONE

- (c) The Contractor shall provide a minimum 10-year warranty in favour of the Principal.

- (d) The following shall be warranted:
- i. Equipment shall remain fit for the intended purpose and comply with the relevant sport governing body requirements (current at time of installation) for the duration of the warranty period.
  - ii. Faults due to poor workmanship in the manufacture and / or installation of the equipment.

# **Attachment – CMDG Construction Specifications**